



NASDAQ GLIMPSE 3.1f

1 Overview

A complement to the NASDAQ TotalView-ITCH real-time data feed product, NASDAQ GLIMPSE 3.1f is a point-to-point data feed connection that provides direct data feed customers with the current state of the NASDAQ Single Book with full market participant attribution. NASDAQ GLIMPSE 3.1f uses the same Add Order messages as the TotalView-ITCH 3.1f data feed product.

2 Architecture

NASDAQ GLIMPSE 3.1f is a point-to-point data feed product comprised of a series of sequenced messages. Each message is variable in length based on the message type and is composed of non-control ASCII bytes. The messages that make up the GLIMPSE 3.1f protocol are typically delivered using a higher level protocol that takes care of sequencing and delivery guarantees.

NASDAQ currently offers the GLIMPSE 3.1f data feed in the [SoupTCP3.0](#) protocol option only. Please note that GLIMPSE users must login to SoupTCP3.0 for sequence 1 to correctly receive data.

3 Data Types

All numeric fields are composed of a string of ASCII coded digits, right justified and space filled on the left.

All alpha fields are left justified and padded on the right with spaces.

Prices are given in decimal format with 6 whole number places followed by 4 decimal digits. The whole number portion is padded on the left with spaces; the decimal portion is padded on the right with zeros. The decimal point is implied by position; it does not appear inside the price field.

4 Message Formats

NASDAQ GLIMPSE firms will receive Add Order messages for all the orders on the NASDAQ Single Book system at the time of login request. NASDAQ GLIMPSE 3.1f uses the same Order message formats as the TotalView-ITCH 3.1f data feed.

At the end of the order spin, GLIMPSE will send a Snapshot message to denote where firms should begin processing real-time updates via the NASDAQ TotalView-ITCH product. .

Please note that NASDAQ GLIMPSE product does not contain Symbol Directory or Trading Action messages. If needed, firms may download the [Symbol Directory file](#) for NASDAQ and other exchange listed issues from the NASDAQ OMX Trader website. Firms may also obtain a list of [Current Trading Halts](#) for NASDAQ and other exchange listed issues from the Market Regulation section of the NASDAQ OMX Trader website.

4.1 Time Messages

For bandwidth efficiency reasons, NASDAQ uses separate time stamp messages for the NASDAQ GLIMPSE 3.1f and NASDAQ TotalView-ITCH 3.1f products. The messages will be sequenced so that data feed recipients can apply the time stamps properly. The time stamp is broken into two separate messages: seconds and milliseconds.

4.1.1 Seconds Message

This message is sent every second for which at least one NASDAQ GLIMPSE 3.1f / NASDAQ TotalView-ITCH 3.1f message is being generated.

SECONDS MESSAGE				
Name	Offset	Length	Value	Notes
Message Type	0	1	"T"	Seconds Message.
Second	1	5	Numeric	Number of second since midnight.

4.1.2 Milliseconds Message

This message is sent every millisecond for which at least one NASDAQ GLIMPSE 3.1f / NASDAQ TotalView-ITCH 3.1f message is being generated.

MILLISECONDS MESSAGE				
Name	Offset	Length	Value	Notes
Message Type	0	1	"M"	Milliseconds Message.
Millisecond	1	3	Numeric	Number of milliseconds since last second.

4.2 System Event Message

NASDAQ GLIMPSE product is designed to support the same system event messages as the NASDAQ TotalView-ITCH feed. On GLIMPSE, NASDAQ will send out system event messages for those TotalView-ITCH events that occurred up to the time of the GLIMPSE snapshot request.

SYSTEM EVENT MESSAGE				
Name	Offset	Length	Value	Notes
Message Type	0	1	"S"	System Event Message.
Event Code	1	1	Alpha	See System Event Codes below.

SYSTEM EVENT CODES – DAILY	
Code	Explanation
"O"	<i>Start of Messages.</i> Outside of time stamp messages, the start of day message is the first message sent in any trading day.
"S"	<i>Start of System hours.</i> This message indicates that NASDAQ is open and ready to start accepting orders.
"Q"	<i>Start of Market hours.</i> This message is intended to indicate that Market Hours orders are available for execution.
"M"	<i>End of Market hours.</i> This message is intended to indicate that Market Hours orders are no longer available for execution.
"E"	<i>End of System hours.</i> It indicates that NASDAQ is now closed and will not accept any new orders today. It is still possible to receive Broken Trade messages and Order Delete messages after the End of Day.
"C"	<i>End of Messages.</i> This is always the last message sent in any trading day.

SYSTEM EVENT CODES – AS NEEDED	
Code	Explanation
"A"	<i>Emergency Market Condition – Halt:</i> This message is sent to inform NASDAQ market participants that the EMC is in effect. No trading is allowed during the EMC.
"R"	<i>Emergency Market Condition – Quote Only Period:</i> This message is sent to inform NASDAQ market participants that the EMC quotation only period is in effect.
"B"	<i>Emergency Market Condition – Resumption:</i> This message is sent to inform NASDAQ market participants that EMC is no longer in effect.

4.3 Add Order Message

An Add Order Message indicates that a new order has been accepted by the NASDAQ system and was added to the displayable book. The message includes a day-unique Order Reference Number used by NASDAQ to track the order.

For GLIMPSE 3.1f / TotalView-ITCH 3.1f, NASDAQ supports two variations of the Add Order message format.

4.3.1 Add Order – No MPID Attribution

This message will be generated for unattributed orders in the NASDAQ Single Book system. (Note: If a firm wants to display a MPID for unattributed orders, NASDAQ recommends that it use the MPID of "NSDQ".)

Please note that the Display Indicator will reflect the value for the order at the time of the NASDAQ GLIMPSE snapshot. Please note that the "S" display indicator is only valid for up to 50 milliseconds from the time that the order was generated or replaced.

ADD ORDER MESSAGE				
Name	Offset	Length	Value	Notes
Message Type	0	1	"A"	Add Order Message.
Order Reference Number	1	12	Numeric	The unique reference number assigned to the new order.
Buy/Sell Indicator	13	1	Alpha	The type of order being added. "B" = buy order. "S" = sell order.
Shares	14	6	Numeric	The total number of shares associated with the order being added to the book.
Stock	20	6	Alpha	The security symbol for which the order is being added.
Price	26	10	Numeric	The display price of the new order. Refer to Data Types for field processing notes.

Display	36	1	Alpha	Indicates the Display code for the order at the time of the GLIMPSE dissemination snapshot. “Y” = Displayable order type on all NASDAQ data feeds “S” = Flash order displayable on TotalView-ITCH only
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4.3.2 Add Order – MPID Attribution

This message will be generated for attributed orders and quotations entered into the NASDAQ Single Book system.

ADD ORDER – MPID ATTRIBUTION MESSAGE				
Name	Offset	Length	Value	Notes
Message Type	0	1	“F”	Add Order Message.
Order Reference Number	1	12	Numeric	The unique reference number assigned to the new order.
Buy/Sell Indicator	13	1	Alpha	The type of order being added. “B” = buy order. “S” = sell order.
Shares	14	6	Numeric	The total number of shares associated with the order being added to the book.
Stock	20	6	Alpha	The security symbol for which the order is being added.
Price	26	10	Numeric	The display price of the new order. Refer to Data Types for field processing notes.
Attribution	36	4	Alpha	NASDAQ market participant identifier associated with the entered order.

4.4 Snapshot message

The Snapshot message reflects the NASDAQ TotalView-ITCH 3.1f sequence number at the time that the NASDAQ GLIMPSE 3.1f spin was requested.

To maintain a real-time order display, firms should begin to process real-time NASDAQ TotalView-ITCH 3.1f messages beginning with the sequence number stated in this GLIMPSE snapshot message.

END OF SNAPSHOT MESSAGE				
Name	Offset	Length	Value	Notes
Message Type	0	1	“G”	End of Snapshot message

TotalView-ITCH 3.0 Sequence number	1	20	Numeric	TotalView-ITCH 3.1f sequence number when the GLIMPSE 3.1f snapshot was taken. To keep the order book current, firms should process real-time TotalView-ITCH 3.1f messages beginning with the message sequence number reflected in this snapshot message.
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5 Support

- o For general product support for NASDAQ data feeds, please contact NASDAQ OMX Global Data Products at +1 301 978 5307 or dataproductions@nasdaqomx.com.
- o For technical support for NASDAQ data feeds, please contact NASDAQ OMX Development Support at devsupport@nasdaqomx.com.

Appendix A

Documentation Revision Control Log

July 20, 2009: NASDAQ GLIMPSE Version 3.1f

NASDAQ releases GLIMPSE 3.1f specifications to the public. The NASDAQ GLIMPSE 3.1f data formats include a new "Display" field in the Add Order – Anonymous data format.

May 20, 2009: NASDAQ GLIMPSE Version 3.10

NASDAQ adds a statement about the user logon procedure for GLIMPSE in section 2.

April 28, 2009: NASDAQ GLIMPSE Version 3.10

NASDAQ releases initial GLIMPSE 3.10 specifications to public.